Exponential Functions and Their Graphs

ANSWER KEY

Section 1: Base e and Exponential Growth/Decay Curves

1 For $f(x) = a \cdot b^x$ with $a > 0$, under what condition does $f(x)$ represent exponential growth?

a. $b > 1$ ✓

b. $0 < b < 1$

c. $b < 0$

d. $b = 1$

Exponential growth occurs when the base $b > 1$. If $0 < b < 1$, it represents exponential decay.

2 Find the horizontal asymptote and y-intercept of $f(x) = 3 \cdot 2^x - 5$.

a. **Asymptote:** $y = -5$**, y-intercept:** $(0, -2)$ ✓

b. Asymptote: $y = 0$, y-intercept: $(0, 3)$

c. Asymptote: $y = -5$, y-intercept: $(0, -5)$

d. Asymptote: $x = 0$, y-intercept: $(0, -2)$

As $x \to -\infty$, $2^x \to 0$, so $y \to -5$. At $x = 0$: $f(0) = 3(1) - 5 = -2$.

3 Determine the range of $g(x) = -2 \cdot e^x + 4$.

a. $(-\infty, 4)$ ✓

b. $[4, \infty)$

c. $(4, \infty)$

d. $(-\infty, -2)$

Since $e^x > 0$, $-2e^x < 0$. Adding 4 gives $g(x) < 4$. The range is $(-\infty, 4)$.

Section 2: Transformations of Exponential Functions

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4 Describe the transformations needed to obtain $g(x) = 2^{x-3} + 1$ from $f(x) = 2^x$.

a. **Shift right by 3 units and up by 1 unit ✓**

b. Shift left by 3 units and up by 1 unit

c. Shift right by 3 units and down by 1 unit

d. Vertical stretch by 3 and up by 1

Replacing x by $x - 3$ shifts right 3; adding 1 shifts up 1.

5 What is the domain of any function of the form $f(x) = a \cdot b^{x-h} + k$ $(b > 0, b \neq 1)$?

a. $(-\infty, \infty)$ ✓

b. $[0, \infty)$

c. (k, ∞)

d. $[h, \infty)$

Exponential functions are defined for every real number x, so the domain is always all real numbers $(-\infty, \infty)$.

6 Which transformation reflects the graph of $f(x) = e^x$ across the y-axis?

a. $g(x) = e^{-x}$ ✓

b. $g(x) = -e^x$

c. $g(x) = -e^{-x}$

d. $g(x) = e^x - 1$

Replacing x with $-x$ reflects the graph across the y-axis.

Section 3: Compound Interest Formulas

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7 State the formula for an initial principal P compounded continuously at annual rate r for t years.

a. $A = Pe^{rt}$ ✓

b. $A = P(1 + r)^t$

c. $A = P(1 + \frac{r}{n})^{nt}$

d. $A = Prt$

Continuous compounding uses the natural base: $A = Pe^{rt}$.

8 If 1000 dollars is invested at 6% compounded monthly, what is the balance after 5 years?

a. $1000(1 + \frac{0.06}{12})^{60} \approx 1348.85$ dollars ✓

b. 1300 dollars

c. 1349.86 dollars

d. 1338.23 dollars

$A = 1000(1 + 0.005)^{60} = 1000(1.005)^{60} \approx 1348.85$ dollars.

9 If 2000 dollars is invested at 5% compounded continuously, find the value after 10 years.

a. $2000e^{0.5} \approx 3297.44$ dollars ✓

b. 3000 dollars

c. 3257.79 dollars

d. 4000 dollars

$A = 2000e^{(0.05)(10)} = 2000e^{0.5} \approx 2000(1.64872) \approx 3297.44$ dollars.

Section 4: The Number e as a Limit

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10 State the limit definition of Euler's number e.

a. $e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$ ✓

b. $e = \lim_{n\to 0} \left(1 + \frac{1}{n}\right)^n$

c. $e = \lim_{n\to\infty} (1 + n)^{1/n}$

d. $e = \sum_{n=1}^{\infty} \frac{1}{n}$

Euler's number $e \approx 2.71828$ is defined by $\lim_{n\to\infty} (1 + 1/n)^n$.

11 Evaluate $\lim_{n\to\infty} \left(1 + \frac{3}{n}\right)^n$.

a. e^3 ✓

b. $3e$

c. $e^{1/3}$

d. 3

Using $\lim_{n\to\infty} (1 + x/n)^n = e^x$ with $x = 3$ gives e^3.

12 Why is continuous compounding optimal compared to discrete compounding at the same nominal rate?

a. **As $n \to \infty$, the compounding factor $(1 + r/n)^{nt} \to e^{rt}$, which achieves the theoretical maximum return** ✓

b. It avoids paying bank fees

c. The interest rate increases

d. It doubles every year

Compounding frequency increases accumulation monotonically, reaching its mathematical supremum in the limit $n \to \infty$ with e^{rt}.

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